Ramesh's Homelab & Media Server Documentation

Overview

This document describes the complete homelab environment maintained by Ramesh.

The environment is built around a Raspberry Pi 3B running CasaOS and serves multiple purposes:

- Media automation
- Media streaming
- Network-wide ad blocking
- Service monitoring
- Secure remote access
- Portfolio hosting
- Real-time system monitoring APIs
- Learning platform for self-hosting and software engineering

---

Owner

Name: Ramesh

Role: Software Engineer (.NET / WPF / ASP.NET)

Experience: 5+ Years

Primary Objectives:

1. Self-host services on low-power hardware
2. Automate Tamil movie acquisition and management
3. Learn DevOps and infrastructure concepts
4. Build portfolio projects using real production-like data
5. Create a central platform for experimentation and automation

---

Hardware

Primary Server

Raspberry Pi 3B

Purpose:

- Main homelab server
- Runs 24/7

Responsibilities:

- Hosts all self-hosted services
- Runs media automation stack
- Provides monitoring APIs
- Serves portfolio integrations

Operating System

CasaOS

Purpose:

Container and application management platform.

Benefits:

- Easy service deployment
- Web-based management
- Docker container support
- Centralized application dashboard

High-Level Architecture

```
                 Internet
                    │
                    ▼

             Portfolio Website
                    │
```

```
                       ▼

                  Custom APIs
                       │
                       ▼

               Raspberry Pi 3B
                       │
 ┌──────────────┬──────┼──────────┬──────────┐
 │              │      │          │          │
 ▼              ▼      ▼          ▼          ▼

CasaOS       Pi-hole   Uptime Kuma   Tailscale   Media Stack
```

Core Infrastructure Services

1. Tailscale

Purpose:

Secure remote access to the homelab.

Capabilities:

- Access Raspberry Pi remotely
- Access CasaOS dashboard
- Access Emby
- Access Pi-hole
- Access monitoring dashboards

Benefits:

- No port forwarding
- Encrypted communication
- Private VPN network

Usage:

```
Laptop / Mobile
       │
       ▼
   Tailscale
       │
```

```
      ▼
Raspberry Pi
```

---

2. Pi-hole

Purpose:

Network-wide ad and tracker blocking.

Functions:

- DNS sinkhole
- Ad blocking
- Tracker blocking
- DNS analytics

Benefits:

- Faster browsing
- Cleaner web experience
- Reduced tracking

Flow:

```
Device
   │
   ▼
Pi-hole
   │
   ▼
Internet DNS
```

---

3. Uptime Kuma

Purpose:

Monitor all hosted services.

Monitored Targets:

- Raspberry Pi
- CasaOS

- Emby
- Radarr
- Jackett
- Pi-hole
- Portfolio APIs
- Website endpoints

Features:

- Uptime monitoring
- Downtime alerts
- Service status dashboard
- Historical availability

Media Automation Stack

Goal:

Automated Tamil movie discovery with manual approval before downloading.

---

1. Radarr

Purpose:

Movie management and automation.

Responsibilities:

- Track wanted movies
- Search releases
- Manage metadata
- Coordinate downloads

Root Folder:

```
/mnt/media/movies/tamil
```

---

2. Jackett

Purpose:

Torrent indexer aggregation.

Responsibilities:

- Connect to multiple torrent sites
- Provide Torznab feeds to Radarr

Connection Type:

```
Torznab
```

---

3. Telegram Bot

Purpose:

Manual approval workflow.

Process:

```
Movie Found
      │
      ▼
Telegram Alert
      │
      ▼
Approve / Reject
      │
      ▼
Download Starts
```

Benefits:

- Avoid fake torrents
- Avoid CAM releases
- Quality control
- Language verification

---

4. Download Client

Transmission

Primary lightweight torrent client.

Benefits:

- Raspberry Pi friendly
- Low memory usage
- Reliable

Default Port:

```
9091
```

Alternative Discussed:

- qBittorrent

---

5. Emby

Purpose:

Media server and streaming platform.

Library Path:

```
/mnt/media/movies/tamil
```

Features:

- Metadata management
- Automatic library refresh
- Streaming across devices

Flow:

```
Download Complete
        │
        ▼
Emby Detects File
```

```
       │
       ▼
Metadata Downloaded
       │
       ▼
Movie Available
```

Media Workflow

```
Movie Added To Radarr
          │
          ▼
Jackett Searches Indexers
          │
          ▼
Torrent Found
          │
          ▼
Telegram Notification
          │
          ▼
Manual Approval
          │
          ▼
Transmission Download
          │
          ▼
Stored In Tamil Library
          │
          ▼
Emby Refresh
          │
          ▼
Ready To Stream
```

Storage Layout

```
/mnt/media

└── movies
```

```
    └── tamil
        ├── Movie 1
        ├── Movie 2
        └── Movie 3
```

Used By:

- Radarr
- Transmission
- Emby

Portfolio Website

Purpose

Professional portfolio website showcasing:

- Experience
- Projects
- Technical skills
- Live server statistics

Live System Dashboard

The portfolio site consumes APIs hosted by the Raspberry Pi.

Displayed Metrics May Include:

- Server uptime
- CPU usage
- RAM usage
- Storage utilization
- Active services
- Media library statistics
- Health status

Custom Monitoring APIs

Purpose:

Expose real-time homelab information.

Architecture:

```
Portfolio Website
        │
        ▼
Custom APIs
        │
        ▼
Raspberry Pi Services
```

Example Endpoints:

```
/api/system
/api/cpu
/api/memory
/api/storage
/api/uptime
/api/services
```

Example Response:

```
{
  "cpu": 32,
  "memory": 58,
  "disk": 71,
  "uptime": "14 days"
}
```

Security Model

Internal Access

Protected through:

- Tailscale

Public Access

Limited to:

- Portfolio Website
- Selected APIs

Security Goals:

- Avoid exposing infrastructure
- Encrypt remote access
- Minimize attack surface

Monitoring Strategy

Uptime Kuma

Monitors:

- Infrastructure
- Applications
- APIs

Portfolio Dashboard

Displays:

- Real-time system metrics

Pi-hole

Monitors:

- DNS activity
- Ad blocking statistics

Design Principles

Low-Power Computing

Hardware:

- Raspberry Pi 3B

Goal:

Maximum functionality with minimal power consumption.

Self-Hosting

Preference for:

- Local ownership
- Open-source software
- Privacy-first solutions

Automation With Human Approval

Downloads should not start automatically.

Reason:

- Quality assurance
- Language verification
- Torrent validation

Learning Platform

Homelab is used to gain experience in:

- Linux
- Docker
- CasaOS
- Networking
- DNS
- Monitoring
- APIs

- Automation
- System Administration

Future Enhancements

Planned

Sonarr

Tamil TV show automation.

Overseerr / Jellyseerr

Media request portal.

Telegram Inline Controls

Approve downloads directly from chat.

Automatic Torrent Cleanup

Remove completed torrents after seeding targets.

Advanced Metrics

Expose more system statistics through APIs.

Additional Services

Potential future self-hosted applications running under CasaOS.

Complete Technology Stack

```
Hardware
--------
Raspberry Pi 3B

Platform
--------
CasaOS

Remote Access
```

```
-------------
Tailscale

DNS Filtering
-------------
Pi-hole

Monitoring
----------
Uptime Kuma

Media Server
------------
Emby

Movie Automation
----------------
Radarr

Indexer Aggregation
-------------------
Jackett

Download Client
---------------
Transmission

Notifications
-------------
Telegram Bot

Custom Development
------------------
Monitoring APIs

Public Presence
---------------
Portfolio Website

Media Type
----------
Tamil Movies

Storage
-------
/mnt/media/movies/tamil
```

Summary

This homelab is a multi-purpose self-hosted environment that combines media automation, infrastructure monitoring, network services, secure remote access, and custom software development on a Raspberry Pi 3B.

It serves as both a production system for daily use and a practical learning environment for software engineering, DevOps, networking, and self-hosting technologies.